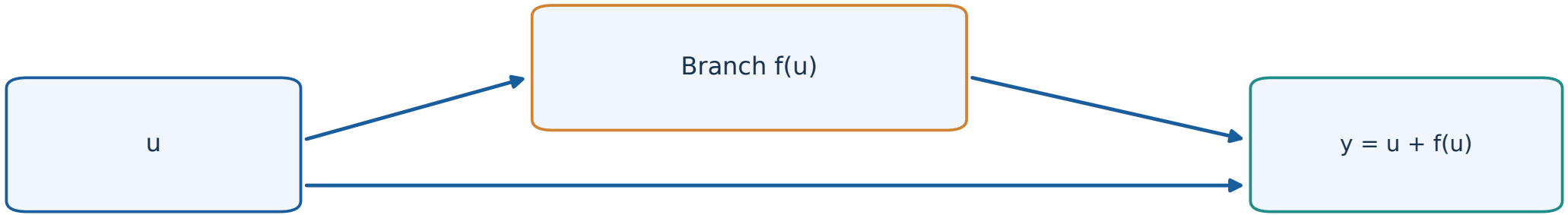
Residual connections retain and transform input dependence

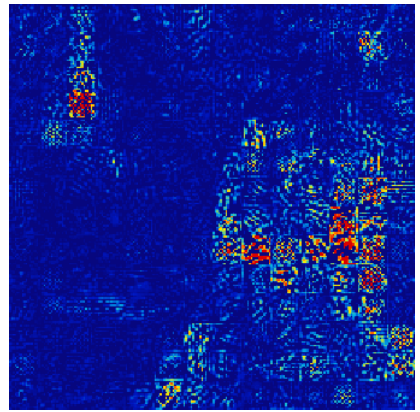


Identity edge uses the output cotangent g_Y

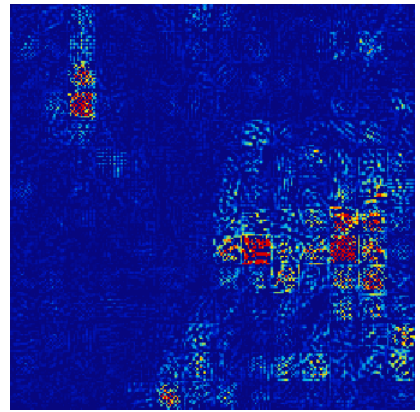
Original image



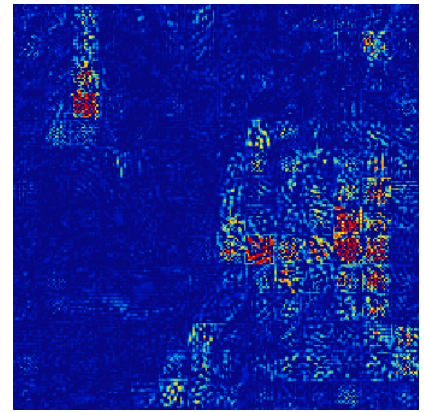
Identity



Attention branch



Residual output



$$E_{\text{output}} = E_{\text{identity}} + E_{\text{branch}}$$